NOT MEASUREMENT SENSITIVE

A-A-51461C
31 May 2000
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

TEST KIT, TEST STRIPS AND COLOR CHART, ANTIFREEZE, FREEZE POINT AND NITRITE CONCENTRATION

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for test kits suitable for determining the freezing point and reserve alkalinity of ethylene glycol antifreeze solutions as well as both the universal freezing point and nitrite concentration of multi engine type antifreeze solutions used in engine cooling systems.

2. CLASSIFICATION.

Type	Pad 1	Pad 2	Antifreeze Glycol Type
I	Freeze Point	Reserve Alkalinity	Ethylene Glycol Only
II	Universal Freeze Point	Nitrite Concentration	Ethylene Glycol and Propylene Glycol

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/IE, Warren, MI 48397-5000.

AMSC N/A FSC 6630

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3. SALIENT CHARACTERISTICS.

3.1 Materials. The test kit both type I and type II shall contain a plastic strip with two reagent pads at the distal end. The test kit will be supplied with instructions and color chart(s) for interpreting the color reactions on each reagent pad as a result of testing a cooling solution in accordance with the manufacturer's instructions. Each test kit containing fifty test strips, as specified, constitutes one unit.

3.2 Requirements.

3.2.1 Type I. Type I test kits shall indicate the freezing point and reserve alkalinity of ethylene glycol antifreeze solutions by dipping the distal end into the solution and then matching the reaction on each reagent pad with the appropriate color chart. Type I shall have one reagent pad on the test strip react to various concentrations of antifreeze protection by changing to a color that is indicative of its freezing point level within a range that includes -51 degrees Celsius (°C) to -12°C (-60 degrees Fahrenheit (°F) to +10°F). Type I shall have a second reagent pad on the test strip react to various concentrations of alkalinity by changing to a color that is indicative of its reserve alkalinity. The reserve alkalinity range shall include 5 milliliters (mL) to 20 mL.

3.2.2 Type II. Type II test kits shall contain a test strip suitable for determining both the universal freezing point and nitrite concentration of both ethylene glycol and propylene glycol based antifreeze solutions. Type II shall indicate the universal freezing point and nitrite concentration of an antifreeze solution by dipping the distal end into the solution and then matching the reaction on each reagent pad with the appropriate color chart. Type II shall have one reagent pad on the test strip react to various concentrations of ethylene glycol and propylene glycol antifreeze protection by changing to a color that is indicative of its universal freeze point level within a range that includes -51°C to -12°C. Type II shall have a second reagent pad on the test strip react to various concentrations of sodium nitrite in antifreeze solutions by changing to a color that is indicative of its nitrite level within a range that includes 0 parts per million (ppm) to 4000 ppm as nitrite ion.

3.3 Identification and markings. Identification and markings shall be permanent and legible and shall include, as a minimum, the manufacturer’s identification code (CAGE), the contract number, and the Part or Identification Number (PIN) (see 7.2).

4. REGULATORY REQUIREMENTS.

4.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspections and testing. Inspection and testing shall be conducted in accordance with commercial practice to determine compliance with the requirements of this commercial item description.

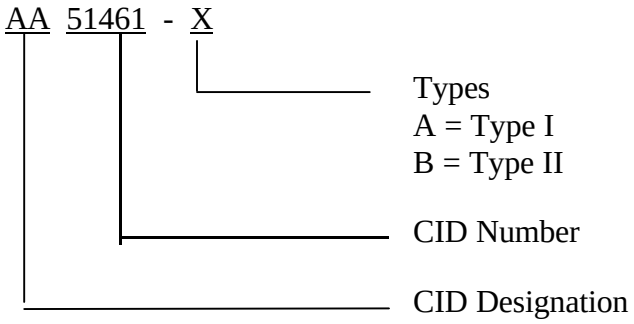
6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES. (This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

7.1 Universal freeze point data. Ethylene glycol based antifreeze solutions and propylene glycol based antifreeze solutions have slightly different freezing points at the same glycol concentration. The following table lists expected universal freeze point values based on concentration of ethylene glycol and/or propylene glycol.

Concentration of glycol, % by volume	Universal freezing point of antifreeze, °C	°F
25	-12	+10
33	-18	0
40	-23	-10
50	-34	-30
60	-51	-60

7.2 Part or Identification Number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. The PINs to be used for test kits acquired to this CID are created as follows:



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7.3 Addresses for obtaining copies of referenced documents.

7.3.1 Government publications. The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.4 Ordering data. Acquisition documents must specify the following:

a. Title, number, date of this CID and CID PIN.
b. Issue of the Department of Defense Index of Specifications and Standards (DoDISS) to be cited in the solicitation.
c. Selection of applicable packaging requirements.

7.5 Keywords.

Ethylene glycol
Reagent pads

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - MS

Review Activities:
DLA - DM
DMSB - MB

CIVIL AGENCY COORDINATION ACTIVITY:
VA-OSS

Preparing Activity:
Army - AT

(Project 6630-0579)

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